Free Worldwide Elevation Data

Commercial-use guide for building a planet-scale game in Unreal Engine

Can you use the data in a paid game?

Yes. The four datasets below are generally available without a terrain-data licence fee for commercial game development. However, their terms differ, and you should verify the current terms for the exact product, version and source files you download—especially if you plan to redistribute raw or lightly processed elevation files.

Dataset	Commercial use	Main considerations	Official source
NASA SRTM	Generally free for commercial use	Global land coverage is roughly 60°N to 56°S. Check the specific product terms and acknowledge the source where appropriate.	NASA SRTM
Copernicus DEM	Free commercial use under its applicable licence	GLO-30 and GLO-90 provide worldwide land coverage. Follow the applicable attribution requirements.	Copernicus DEM
NOAA ETOPO 2022	Yes; CC0 public-domain dedication	Global land and ocean-floor topography. Attribution is recommended even though CC0 does not require it.	NOAA ETOPO
GEBCO	Free commercial use under its terms	Global land and seafloor grid. Acknowledge GEBCO and do not imply official endorsement.	GEBCO grids

Recommended combination

For detailed land: Copernicus DEM GLO-30 is a strong starting point. **For a complete planet:** combine land elevation with NOAA ETOPO 2022 for global ocean-floor depth. NASA SRTM is another useful land-elevation source. GEBCO is an alternative global land-and-seafloor grid.

Attribution example for Copernicus DEM

The applicable notice depends on the Copernicus product and licence version. A commonly published acknowledgement is:

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Important licensing distinction

Using elevation data to create terrain inside a commercial game is different from republishing the original dataset. Check the licence for each downloaded file, retain source/attribution records, and review any restrictions on redistribution, sublicensing, or third-party components. This guide is practical information, not legal advice.

Unreal Engine implementation notes

- **Do not expect one giant Landscape:** Unreal Landscapes are bounded terrain sections.
- **Use a spherical planet:** for a globe-scale game, build an Earth sphere or cube-sphere with tiled terrain.
- **Stream by distance:** load high-resolution tiles near the player and lower-resolution terrain farther away.
- **Prepare the data:** use QGIS or GDAL to mosaic, reproject, crop and export suitable heightmaps.
- **Choose resolution strategically:** 30 m data is useful for broad terrain but does not contain building- or street-level detail.

Official dataset links are included above. Licensing and dataset availability can change; check the provider's current terms before release.